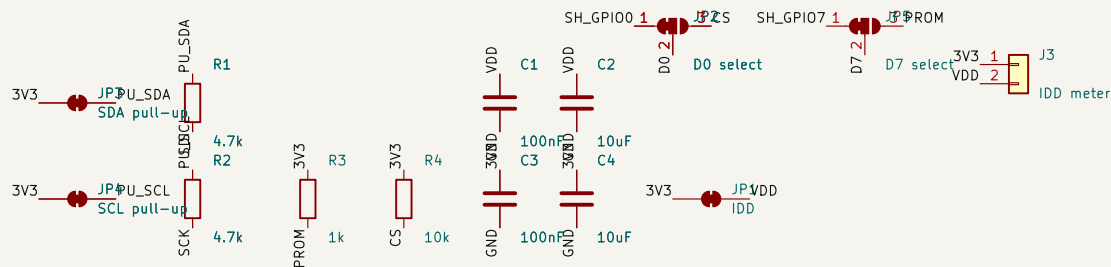


Board B: XIAO ESP32-S3 carrier for a Bosch Shuttle Board 3.0. Shuttle on top, XIAO underneath. 3.3 V only.
 BME690 8x shuttle: D0–D7 (GPIO1, 2, 3, 4, 5, 6, 43, 44) are the chip selects of sensors U1–U8;
 D8 SCK (GPIO7), D9 SDO/MISO (GPIO8), D10 SDI/MOSI (GPIO9). The XIAO Sense's microSD card shares
 D8–D10 (its own CS is GPIO21), so on the 8x board the card and the sensors take turns on one SPI bus.

P1	P2
VDD <u>1</u>	VDD <u>1</u>
3V3 <u>2</u>	CS <u>11</u>
GND <u>3</u>	SCK <u>12</u>
SH_GPIO0 <u>4</u>	SDO <u>13</u>
GPIO1 <u>5</u>	SDI <u>14</u>
GPIO2 <u>6</u>	GPIO4 <u>15</u>
GPIO3 <u>7</u>	GPIO5 <u>16</u>
	GPIO6 <u>17</u>
	SH_GPIO7 <u>18</u>
	PROM <u>19</u>
	PROM_RW <u>20</u>
	Shuttle <u>21</u>
	3.0 sockets

D0	<u>1</u>	D0/GPIO1	U1	14
GPI01	<u>2</u>	D1/GPIO2		13 GND
GPI02	<u>3</u>	D2/GPIO3	3V3	12 3V3
GPI03	<u>4</u>	D3/GPIO4		11 DSI
GPI04	<u>5</u>	D4/GPIO5		10 SDC
GPI05	<u>6</u>	D5/GPIO6		9 SCK
GPI06	<u>7</u>	D6/GPIO7		8 D7
		D7/GPIO8	XIAO	ESP32C3



Selector jumpers, set for the 8x shuttle as made (pads 1–2 joined):
 JP2: D0 → shuttle GPIO0 (8x: sensor U1 CS). Cut 1–2, bridge 2–3; D0 → shuttle CS (single-sensor shuttles).
 JP5: D7 → shuttle GPIO7 (8x: sensor U8 CS). Cut 1–2, bridge 2–3; D7 → PROM_RW (shuttle ID chip).
 JP1 IDD: cut and put a meter across J3 to measure the sensor VDD current.
 JP3 / JP4: 4.7k I2C pull-ups on SDA / SCL; cut to remove.

R3 1k: PROM_RW pull-up for the DS28E05 1-Wire ID chip (datasheet: 300–1500 ohm at 3.3 V, overdrive speed only). R4 10k: CS pull-up so single-sensor shuttles start in I2C mode. VDDIO is tied to the XIAO's 3V3 (its logic level), so there is no TIE jumper on this board.

github.com/leakydata/bme690-toolkit
 Seeded XIAO ESP32-S3 / Sense carrier for a Bosch Shuttle Board 3.0

Sheet: /
 File: board_b.kicad_sch

Title: Board B – XIAO ESP32-S3 shuttle carrier

Size: A4 Date: 2026–10–03

KiCad E.D.A. 10.0.6

Rev: 1

Id: 1/1